

TestVDB: Detecting API Compliance Defects in Vector Database Systems via Contract-Truth Separation

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Abstract

Vector Database Management Systems (VDBMSs) have become critical infrastructure for Large Language Model (LLM) applications, yet their reliability is underexplored: incorrect-behavior defects account for 43% of VDBMS bugs—far exceeding crash/hang bugs (23%)—and current fuzzers such as VDBFuzz rely on crash oracles that cannot detect this majority class. We introduce *API compliance defects*—bugs where a VDBMS silently accepts inputs or produces behaviors that violate its documented contract—as a tractable, previously unsystematized slice of this gap, and present **TestVDB**, the first LLM-driven detector for them. TestVDB auto-derives a contract oracle from official API documentation and applies **Contract-Truth Separation** (CTS): it isolates LLM-generated assertions from a truth layer that falsifies them through maintainer-authority evidence (source grounding, clean reproduction, by-design intent). CTS is necessary because we identify and characterize *contract hallucination propagation*: when an LLM both generates a contract and judges compliance, hallucinated constraints are self-confirmed in the generation–judgment chain—12 of our submissions were judged by maintainers as by-design—the LLM-derived contract was stricter than the true intent. Across five VDBMSs, TestVDB produced 111 candidate issues; maintainers acknowledged 36 (28 fixed, 8 accepted), raising end-to-end precision from 12.9% (single-layer LLM judgment) to 69.2% (5.4×). We state the system’s boundary honestly: it specializes in boundary/validation compliance (75% of yield), is complementary to crash-focused fuzzing, and does not solve the result-correctness oracle (ANN recall, ranking), which remains open.

CCS Concepts

• Software and its engineering; • Information systems;

Keywords

Vector database, API compliance defects, LLM-driven testing, test oracle, contract hallucination, multi-agent verification

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1 Introduction

VDBMSs such as Milvus, Qdrant, Weaviate, Chroma, and MeiliSearch now underpin retrieval-augmented generation, recommendation, and multimodal search. As they enter production at scale, their reliability directly affects downstream decisions. A study of 1,671 bug-fix pull requests across 15 VDBMSs [7] finds that more than

Table 1: Oracle candidates for API compliance defects.

Candidate oracle	Why it fails for compliance defects
Crash (VDBFuzz [6])	Compliance defects do not crash
PoC reproduction	No observable failure signal
Differential testing	No reference implementation for VDBMS APIs
Equivalence transformation	No query form to transform equivalently
LLM	Only flexible semantic candidate—but it both generates the contract and judges, both unreliable

half of all defects are functional failures; a complementary testing roadmap [5] reports that *incorrect-behavior* bugs (43.0%) substantially outnumber crash/hang bugs (23.1%).

Despite this distribution, current VDBMS testing skews toward the minority. VDBFuzz [6], the first dedicated VDBMS fuzzer, uses crash as its oracle and detects only crash/hang bugs, leaving the majority without a practical oracle. The roadmap [5] flags this as the central open challenge: incorrect behavior “poses significant challenges for test oracles,” and approximate vector search renders traditional differential testing unsuitable. Its Future Work asks both for “an oracle for evaluating the correctness of vector search results” and for methods that let LLMs “stay updated with the latest syntax and functionalities.”

We target a tractable slice of this gap: *API compliance defects*—bugs where the VDBMS silently accepts an input or produces a behavior that violates its documented contract (e.g., accepting `nprobe=0`, silently normalizing `shardsNum=-1`, or returning success where the docs prescribe rejection). These bugs do not crash, but they corrupt query semantics, lower recall, and expand the attack surface. Crucially, they admit an oracle where general result-correctness does not: the API contract itself.

Why an LLM, and why that is not enough. Compliance defects produce no observable failure signal, and VDBMS APIs have no widely-agreed reference implementation for differential testing. By exclusion (Table 1), the only candidate oracle capable of flexible semantic judgment over an undocumented boundary is an LLM. But using an LLM as the oracle introduces two unreliable roles: it *generates* the contract (from docs) and it *judges* compliance—and both can hallucinate.

Core reversal: the oracle itself may be wrong. The naive design—an LLM extracts a contract from docs, then the same family of models judges compliance against it—is brittle because of *contract hallucination propagation*. We observed a concrete instance: the formalizer “extracted” a complexity requirement from `constant.go` that is in fact a guard against a historical performance regression, not a behavioral constraint; the judge then confirmed violations

against it. Across our submissions, 12 were marked by-design (§4): the LLM-derived contract was stricter than the maintainers’ intent (demanding rejection of idempotent duplicate creates, eventual-consistency staleness, or lenient defaults). When the same model family generates the contract and judges compliance, hallucinated constraints are self-confirmed.

Approach: Contract-Truth Separation. When no mechanical oracle is available, the only usable truth source is *maintainer authority*—source code, prior PRs, by-design intent, issue history. It is scarce, so it must be proxied by an agent. We separate the *assertion layer* (LLM-generated contracts and judgments) from a *truth layer* and use the latter to falsify the former. The truth layer is a dev-reviewer agent that applies counter-evidence along three anchors: clean reproduction, source-grounded verification (the direct counter to hallucination), and threat-model cross-check. This mirrors how a maintainer adjudicates an incoming bug report.

Results. TestVDB produced 111 candidate issues across five VDBMSs; maintainers acknowledged 36 (28 fixed, 8 accepted-open). The dev-reviewer alone removes 80.6% of false positives (FPs) at the Stage-2 layer, and end-to-end precision rises from 12.9% (4/31 on a Milvus ablation) to 69.2% (36/52), a 5.4× improvement. We report the boundary honestly: 75% of yield is boundary/validation compliance; crash bugs are excluded by design (complementary to VDBFuzz); soft result-correctness (approximate nearest neighbor (ANN) recall, ranking) remains open.

Positioning. We respond to VDBFuzz’s generation-side Future Work (LLM-generated diverse API interactions and staying current with evolving APIs). Our judgment side provides a contract oracle that extends detection beyond crash into the API-compliance subset of incorrect behavior. We do *not* claim to solve VDBFuzz’s oracle-for-correctness direction (result correctness of vector search), which remains open [5].

Contributions.

- (1) The **first LLM-driven realization and large-scale empirical study of VDBMS API-compliance defect detection**—on the agenda defined by [5], the first end-to-end realization for non-crash compliance defects, validated at scale (111 issues, 36 acknowledged).
- (2) **Contract-Truth Separation**, a design principle isolating LLM-generated assertions from a maintainer-authority truth layer that falsifies them.
- (3) A **counter-evidence mechanism** (dev-reviewer) with three anchors, removing 80.6% of false positives and lifting precision 5.4×, with its TP-recall limitation (20–60%) reported honestly.
- (4) The **contract hallucination propagation finding**: when an LLM both generates a contract and judges compliance, hallucinated constraints are self-confirmed—observed in 12 by-design cases; the dev-reviewer’s counter-evidence anchors are the mitigation.
- (5) An **exploratory pilot** of a threat-model prior, with honest design limitations; we do not claim a validated effect, and full evaluation is deferred to future work.

Table 2: TestVDB scope projected onto the roadmap symptom taxonomy [5].

Symptom class (share)	TestVDB coverage
Crash/Hang (23.1%)	Out of scope (L1 gate filters; 1 incidental in yield)
Incorrect Behavior (43.0%)	In scope: boundary/validation (75% of yield) + diagnostic + state/logic subsets
Performance (9.3%)	Out of scope
Build (9.7%)	Out of scope

- (6) **Empirical results and an open-source system:** 111 issues across 5 VDBMSs, 36 maintainer acknowledgments; an anonymized artifact is available at <https://anonymous.4open.science/r/testvdb-anon-D644/>.

2 Background and Problem Formulation

2.1 VDBMS Architecture

A VDBMS stores, indexes, and queries dense vector embeddings across a storage engine, a vector index (HNSW, IVF, etc.), a query processor (search, filtered/predicated search, multi-vector search), and client SDKs exposing REST and gRPC APIs [5]. Our targets live at the API surface and the query-processing logic behind it.

2.2 Bug Taxonomy and Our Scope

We adopt the symptom-level taxonomy of [5] as our primary frame (Crash/Hang 23.1%, Incorrect Behavior 43.0%, Performance 9.3%, Build 9.7%) and corroborate it with the larger empirical study [7] (5 symptom categories, 31 fault patterns) used as a fault-pattern vocabulary. Our compliance-dimension axis projects onto the symptom axis as in Table 2: we operate inside Incorrect Behavior, specialize in its boundary/validation subset, and exclude Crash (by design), Performance, and Build.

2.3 API Compliance Defects

An *API compliance defect* is a behavior where, for an input governed by a documented or implementable contract, the VDBMS (i) accepts an input the contract prescribes to reject (*illegal success*), (ii) rejects or mishandles an input the contract prescribes to accept, or (iii) returns a response whose status, payload, or side effects violate the contract (*poor diagnostics / state violation*). The contract is not assumed perfect; Section 4 confronts contract unreliability directly.

2.4 The Oracle Problem

Compliance defects produce no crash and no exception, and—unlike SQL—there are no widely-agreed reference semantics for VDBMS APIs across vendors. Differential testing is unsuitable because APIs diverge and results are approximate [5]. Metamorphic relations [4] serve result correctness but not API-acceptance decisions. The contract itself is the only candidate oracle—but it must be obtained and trusted, which is the core problem we address.

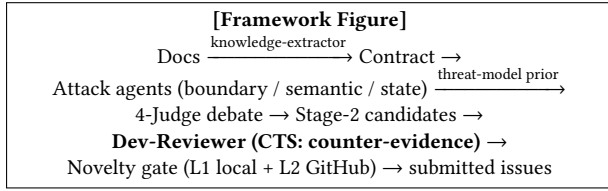


Figure 1: TestVDB pipeline. The dev-reviewer is the core contribution; the threat-model prior spans generation, judgment, and dedup.

3 Approach

3.1 Overview

TestVDB is a five-stage pipeline (Figure 1): (1) contract extraction from API documentation via an LLM knowledge-extractor; (2) attack generation by specialized agents guided by a threat-model prior; (3) a four-judge debate producing Stage-2 candidates; (4) a dev-reviewer that applies Contract-Truth Separation as counter-evidence; (5) a two-layer novelty gate. A threat-model artifact is reused across stages as an experience prior.

3.2 Contract Extraction and the Threat-Model Prior

A knowledge-extractor agent crawls official API documentation and emits structured contracts (parameter domains, enum values, documented limits, status-code semantics) that seed both attack agents and judges. A threat-model artifact encodes recurring by-design intents (idempotency, eventual consistency, lenient defaults) and known blind spots; it is injected as a prior at generation (steering attacks toward non-trivial boundaries), at judgment (as a counter-evidence dictionary for the dev-reviewer), and at dedup. We treat the threat model as an *exploratory pilot* (Section 5.4) rather than a core claim, because our ablation could not cleanly isolate its effect.

3.3 Attack Generation and the Four-Judge Debate

Three attack agents target the compliance dimensions: a *boundary* agent (illegal-success at parameter, enum, and limit edges—the dominant yield), a *semantic* agent (diagnostic-quality violations), and a *state* agent (state/logic inconsistencies). Candidates enter a four-judge debate (evidence/severity/novelty/doc axes) drawing on multi-agent verification ideas [1]. The Stage-2 output of this debate feeds the dev-reviewer.

3.4 Contract-Truth Separation and the Dev-Reviewer

Single-layer LLM judgment is unreliable because the same model family generates the contract and judges compliance. CTS breaks this self-confirmation by introducing an independent truth layer. The dev-reviewer agent proxies maintainer authority and falsifies each Stage-2 candidate along three anchors:

- **Clean reproduction.** Build a minimal reproducer against a live VDBMS of the pinned target version; reject candidates whose

“defect” is a tooling artifact (we withdrew early candidates that misread a missing `rowCount` field as “returns -1”).

- **Source-grounded verification.** Locate the implementation behind the API and check whether the observed behavior is intended—the direct counter to contract hallucination (Section 4).
- **Threat-model cross-check.** Match against known by-design intents before flagging a violation.

The three anchors are complementary, not redundant. Source-grounding is decisive when the code *explicitly* encodes intent—e.g., a name validator that permits a leading underscore, or a parameter with an explicit default—exposing such cases as by-design. But when the source is *silently absent* (no validation for a parameter, or partial concurrency locking), the absence is ambiguous: it may be intended permissiveness or a genuine gap, and an LLM judge defaults to “bug.” These silent-absence cases are exactly what the threat-model anchor (known by-design patterns: idempotency, eventual consistency, lenient defaults) is designed to catch. A controlled retrospective on our historical candidates (§5.3) confirms this boundary empirically: on all 16 historical false positives, source-grounding lifted suppression from 31% to 81% (Milvus 33%→75%, non-Milvus 0%→100%), and the residual misses were uniformly silent-absence cases that require the threat-model anchor.

3.5 Novelty Gate

A two-layer gate prevents re-reporting: L1 checks local history and the threat-model blind-spot set; L2 queries the target repository’s issues and PRs. Candidates surviving both are submitted.

4 Contract Hallucination Propagation

We highlight a phenomenon that, to our knowledge, has not been characterized in LLM-driven testing. When an LLM generates the contract *and* judges compliance, hallucinated constraints are not caught by the judge—they are produced by the same generation-judgment family and thus inherit mutual confirmation. We observed two forms.

(1) **Fabricated provenance.** The formalizer produced a “complexity requirement” attributed to `constant.go`; source grounding showed the constant guards a past performance regression, not a behavioral constraint. The Stage-2 judge nonetheless confirmed violations against it.

(2) **Over-strict intent.** Of 48 submissions adjudicated as bug-or-by-design (excluding 4 wontfix/rejected), 12 (25%) were marked by-design: the contract demanded rejection of behaviors maintainers intended to allow—idempotent duplicate creates returning success, eventual-consistency staleness, lenient naming defaults, and silent enum/empty substitutions.

Formally, let C_{LLM} be the LLM-derived contract and C_{true} the maintainer intent. Single-layer judgment assumes $C_{LLM} = C_{true}$; when $C_{LLM} \supset C_{true}$ (stricter), genuine by-design behaviors are flagged as violations, and the judge—sharing the generator’s bias—does not dissent. The dev-reviewer’s counter-evidence anchors are the mitigation—source-grounding for explicitly-encoded intent, the threat-model anchor for silent-absent by-design patterns—reintroducing an external truth signal (C_{true} proxied by source and history). A quantitative study of hallucination frequency is future

Table 3: Issues submitted and maintainer outcomes (5 VDBMSs). Not-a-bug = By-design + Rejected (maintainer judged not a defect; labels differ across projects). Pending = open bugs awaiting triage. Precision is computed over adjudicated issues (Fixed + Accepted + Not-a-bug = 52): 36/52 = 69.2%. An additional 29 submissions were closed without label or marked duplicate and are excluded.

VDBMS	Submitted	Fixed	Accepted	Not-a-bug	Pending
Milvus	51	14	8	12	0
Qdrant	26	11	0	3	8
Weaviate	30	3	0	1	21
MeiliSearch	3	0	0	0	0
Chroma	1	0	0	0	1
Total	111	28	8	16	30

work; the 12 by-design cases are direct observations of the phenomenon.

5 Evaluation

We address four research questions.

5.1 RQ1: Detection Capability

We ran TestVDB against five VDBMSs, pinning exact target versions. Table 3 reports the yield. The dev-reviewer’s source-grounding fetches version-pinned source via release tags (e.g., /blob/v2.6.17), not live master; the contract’s derivation tag may lag the exact target version (e.g., a v2.6.19 run reused a v2.6.17-derived contract), a drift we flag as a threat to validity (§5.3).

Defect-type distribution. Of the 36 acknowledged true positives (TPs), 27 (75%) are boundary/validation, 3 (8%) diagnostic-quality, 2 (6%) state/logic, 1 (3%) crash, and 3 (8%) result-correctness. The result-correctness cases are instructive: they include COSINE distance returning > 1.0 for identical vectors (a hard mathematical-invariant violation, reproduced on both Milvus and Qdrant), incomplete index results (2 of 25 matching points returned), and a payload filter returning points with a missing field. These are detectable because they violate expressible invariants; soft result-correctness (ANN recall/ranking error) remains open.

Complementarity with VDBFuzz. Our yield is almost entirely non-crash (35/36); VDBFuzz targets crash bugs. The near-zero overlap is by design (our L1 gate filters crash-prone inputs). We frame the two systems as complementary, not competitive.

5.2 RQ2: Case Studies

Contract hallucination (Milvus). The formalizer fabricated a complexity requirement; the dev-reviewer’s source-grounded anchor traced it to `constant.go` and reclassified the candidate, preventing a spurious submission. **Dual-library cosine invariant.** The COSINE distance > 1.0 for identical vectors was independently reproduced on Milvus and Qdrant, confirming a cross-vendor correctness class amenable to contract oracles.

5.3 RQ3: Precision Ablation

On a Milvus v2.6.19 ablation, 1,834 raw candidates collapsed to 31 Stage-2 outputs, of which 4 were true positives—a single-layer LLM-judgment precision of 12.9% (4/31). Adding the dev-reviewer removes 25/31 false positives (80.6%). End-to-end across all five VDBMSs, precision is

$$\text{precision} = \frac{\text{acknowledged}}{\text{acknowledged} + \text{by-design} + \text{rejected}} = \frac{36}{36 + 12 + 4} = 69.2\%$$

a 5.4× improvement over the 12.9% single-layer baseline. We deliberately include *rejected* in the denominator (treating wontfix/not-planned as maintainer-judged false positives) for a conservative estimate.

Controlled retrospective (52 candidates). We re-triaged all 52 maintainer adjudicated candidates under two blind conditions—claim-only (the 4-judge layer) and source-grounded (the dev-reviewer’s source anchor)—with outcomes hidden via label-isolated agents (GLM-5.2). Claim-only suppressed 5/16 FP ($S=31\%$, TP recall $33/36 = 92\%$). Across all 16 FPs (12 Milvus + 4 non-Milvus: Weaviate/Qdrant), adding source-grounding lifted suppression from 5/16 (31%) to 13/16 (81%, 2.6×): Milvus $33\% \rightarrow 75\%$, non-Milvus $0\% \rightarrow 100\%$. The explicit-intent cases (idempotent drop/create, explicitly-allowed naming, DB-level locks, configurable defaults, consistency semantics, dynamicEF sentinel values, non-atomic-by-design writes, unbounded score ranges) were correctly identified as by-design. The residual misses were all *silent-absence* cases (no code-level validation: shardsNum, metricType, empty query vector)—ambiguous signals that require the threat-model anchor (§3.4). This empirically validates the 3-anchor complementarity at scale and across libraries: source-grounding for explicit intent, threat-model for silent absence. Three honesty notes: (i) source-grounding used version-pinned release tags (e.g., /blob/v2.6.17), not master; the derivation tag may lag the exact target version (drift, flagged as a threat); (ii) the 36-TP source-grounding (confirming retained TPs are not flipped, possibly reinstating the 3 claim-only false-suppressed TPs) is completing; the FP-side result here is the complete 16-FP measurement; (iii) this measures judgment-layer precision on a historical candidate pool, not an end-to-end false-positive rate.

5.4 RQ4: Threat-Model Pilot (Exploratory)

We ran a double-blind ablation on Milvus v2.6.17 (control with threat-model cognition vs. experiment without; $n=5$ TP, 7 FP). TP recall moved 20%/60% (control/experiment); FP-correct 86%/57%. We do *not* claim a threat-model effect: blindspot indicators were never populated (so the TP-protection mechanism went untested), the dev-reviewer was unstable on the same candidate across runs (2/2 CONFIRMED/FP), and $n=5$ is below significance. The reliable finding is orthogonal and humbling: the dev-reviewer’s TP recall is only 20–60%—even with counter-evidence, an LLM proxy struggles to recognize genuine bugs, a core limitation of CTS. In concrete terms, CTS suppresses 80.6% of false positives (Milvus ablation, $n=31$) while TP recall sits at only 20–60% (pilot, $n=5$); the latter is small-sample and will be tightened by the full re-triage.

5.5 Threats to Validity

Internal: maintainer acknowledgment is a weak ground truth; rejected/by-design outcomes involve human judgment. *External:* the cross-layer ablation (RQ3) is Milvus-only; cross-library dev-reviewer data is incomplete (Weaviate has 22 open undiagnosed). *Construct:* defect-type classification is title-based and pending per-defect label confirmation. *LLM variance:* dev-reviewer judgments vary across runs; we report ranges, not point estimates.

6 Related Work

VDBMS testing. VDBFuzz [6] is the first dedicated VDBMS fuzzer (crash-oracle-based) and is complementary to this work. The roadmap [5] and the empirical bug study [7] define the agenda and taxonomy we build upon. MeTMAP [4] applies metamorphic testing to false vector matching in LLM-augmented generation. BUZZBEE [8] fuzzes general DBMSs.

Database correctness oracles. NoREC [2] tests SQL via pivoted query synthesis; DDLCheck [3] tests DDL correctness. These assume reference SQL semantics absent in VDBMS APIs.

LLM-based testing and multi-agent debate. LLMs are increasingly used for test generation and oracle enhancement; we use multi-agent debate [1] for generation and judgment but contribute the Contract-Truth Separation layer that addresses the contract-hallucination failure mode of self-confirming LLM oracles.

7 Conclusion and Future Work

We presented TestVDB, the first LLM-driven system for detecting API compliance defects in VDBMSs, built on Contract-Truth Separation. The dev-reviewer counter-evidence mechanism lifts precision 5.4× by falsifying LLM-generated assertions against maintainer authority, and its necessity is motivated by our characterization of contract hallucination propagation. We bound the system honestly: it specializes in boundary/validation compliance, is complementary to crash-focused fuzzing, and does not solve result-correctness oracles. Future work includes completing the source-grounded TP pass of the 52-candidate retrospective (the FP pass reports a 31%→81% suppression lift and surfaces a source-anchor boundary—explicit intent versus silent absence—that empirically validates the 3-anchor dev-reviewer), cross-library ablation, and a quantitative study of contract hallucination frequency.

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